Hidden Technical Debt in Machine Learning Systems

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ABSTRACT

Machine learning offers a fantastically powerful toolkit for building useful complexprediction systems quickly. This paper argues it is dangerous to think ofthese quick wins as coming for free. Using the software engineering frameworkof technical debt, we find it is common to incur massive ongoing maintenancecosts in real-world ML systems. We explore several ML-specific risk factors toaccount for in system design. These include boundary erosion, entanglement,hidden feedback loops, undeclared consumers, data dependencies, configurationissues, changes in the external world, and a variety of system-level anti-patterns.